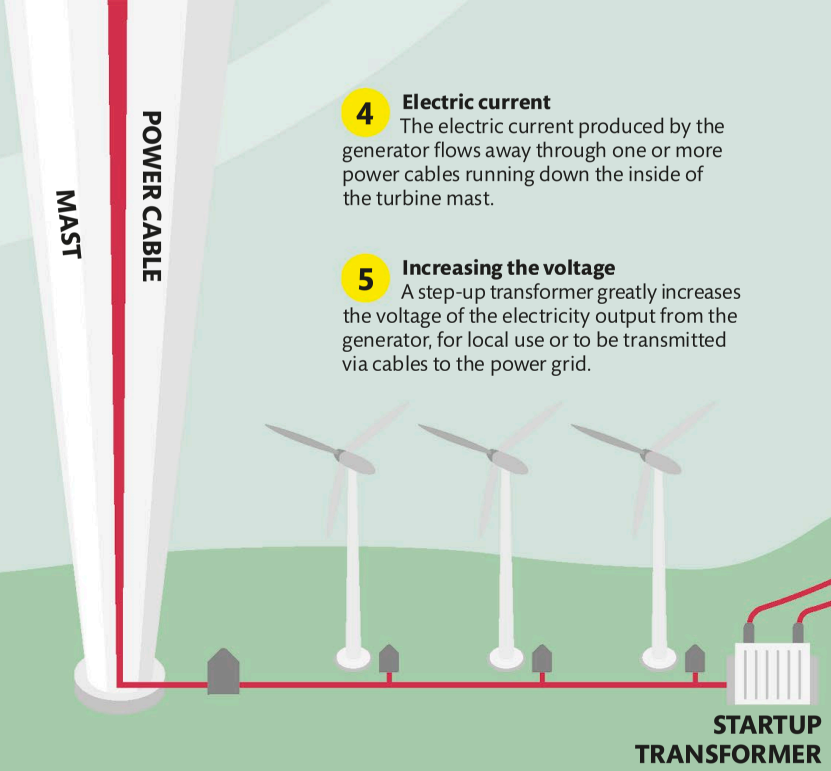
Wind power

For centuries, the power of the wind has been harnessed to propel sailing ships and to drive windmills. Modern wind turbines provide a renewable source of energy by turning the kinetic energy of wind into electrical energy without consuming fossil fuels or emitting greenhouse gases.



4 Electric current
The electric current produced by the generator flows away through one or more power cables running down the inside of the turbine mast.

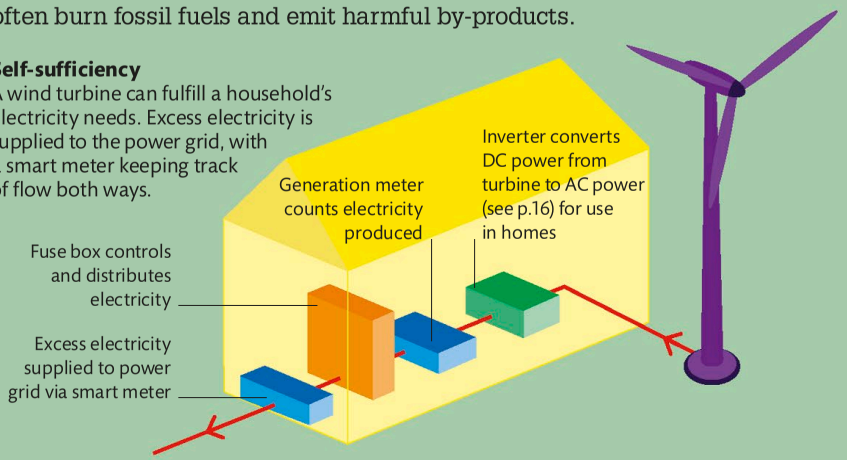
5 Increasing the voltage
A step-up transformer greatly increases the voltage of the electricity output from the generator, for local use or to be transmitted via cables to the power grid.

AN AVERAGE **WIND TURBINE** CAN GENERATE ENOUGH **ELECTRICITY** TO POWER **1,000 HOMES**

Microgeneration

Small-scale renewable energy systems use freestanding or roof-mounted wind turbines to generate electricity, often in conjunction with other sustainable energy sources such as solar thermal collectors to heat water or photovoltaic cells. Together, they reduce the reliance on large, centralized power plants that often burn fossil fuels and emit harmful by-products.

Self-sufficiency
A wind turbine can fulfill a household's electricity needs. Excess electricity is supplied to the power grid, with a smart meter keeping track of flow both ways.



TURBINES AND WILDLIFE

Wind turbine construction can disturb ecosystems at sea and on land, but the most direct threat they pose is to birds and bats. Siting wind farms away from nesting sites and routes used by migrating birds is one solution. Another potential option might be "acoustic lighthouses"—devices placed close to wind turbines that emit loud sounds to warn birds.

